

Lecture 1

The Natural Numbers

- $\mathbb{N}$ is the set of natural numbers.
- $\{1, 2, 3, \dots\}$
- Remark on 0: 0 is considered a natural number in some countries (US, Germany, Russia), but not in other (UK, France, Romance countries)
- Pugh's argument: 0 should be included since counting naturally starts as 1 (such as counting livestock)

History of Math

- Logical deductions starting with Greeks
- Remarks on Ramanujan (he pronounces it according to Jay Cumming's *Math History*, which he recommends)
 - Formulas like $\frac{1}{\pi} = \frac{1}{4} \sum_{n=0}^{\infty} \frac{(6n+1)(\frac{1}{2})^{3n}}{(n!)^3} \frac{1}{4^n}$
 - Hardy getting proofs from Ramanujan
 - Point of this remark: still those who don't fully prescribe to the ideology of proofs

Newton's Infinitesimal vs. Cantor's Infinite

- Cantor's idea of set theory: many turn into 1
 - Modern for his time

Theorem 1

- $\sqrt{2}$ is not rational
 - $\forall \frac{p}{q}, \frac{p^2}{q^2} \neq 2$
 - Proof by contradiction
 - Suppose $r = \frac{p}{q}$ and $\frac{p^2}{q^2} = 2$.

$$\frac{p^2}{q^2} = 2 \Rightarrow p^2 = 2q^2 \Rightarrow 2 \mid p^2 \Rightarrow 4 \mid p^2 \Rightarrow 4 \mid 2q^2 \Rightarrow 2 \mid q^2$$

- Contradiction since $2 \mid p, q$, and they are defined to not share any common factors.
- Question / "something to think about in the shower": Which numbers have rational roots? Does $\sqrt[10]{15}$?

Remark on $\emptyset$

- Cantor disliked: if there is no intersection between 2 sets, it can't be spoken of
- $\mathbb{N} \cup \{0\}$ makes sense, but $\mathbb{N} \cup 0$ doesn't necessarily
 - Reflects different types in Lean (according to what Pugh saw online)

Real Numbers

- $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$
 - $\mathbb{Z}$ introduces subtraction
 - $\mathbb{Q}$ introduces division
- Idea of getting $\mathbb{R}$ is to complete $\mathbb{Q}$
- $\mathbb{R}$ must satisfy:
 - Field
 - Ordered ($<$ makes sense)
 - LUB (Least Upper Bound) Property
 - Apparently very important, will show in some way or another in many proofs
 - $\forall S \in \mathbb{R} (S \neq \emptyset \wedge S \text{ is bounded above} \rightarrow \exists \text{ LUB for } S)$

- In other words, $\exists$ real number x s.t. $\forall s \in S (s \leq x)$.
- The problem
 - Consider a number line of rationals
 - Many “holes” at places where irrational values would be
 - Sorry idk how to draw diagrams
 - Abbott proves real numbers exist at the end of the book (BAD!!)
- How to complete?
 - Dedekind cuts
 - Uses set theory, the simplest method
 - Cauchy sequences
 - Set of all Cauchy sequences, converge to reals
 - Requires more complex math
 - Decimal expansions
 - Difficulties in showing $\mathbb{R}$ is a field (infinite carrying for addition, more issues for multiplication)
- Dedekind Cuts
 - Definition: A cut is a pair of nonempty subsets A and B of $\mathbb{Q}$. $A \cup B = \mathbb{Q}$ and $A \cap B = \emptyset$.
 $\forall a \in A, \forall b \in B, a < b$. A contains no largest element
 - Once again I can't draw the diagram
 - Think of cutting a number line, get intuitive notation of $A \mid B$.
 - Can also refer to a cut by A , since $B = A^C$
 - If $a \in A$, then all elements $< a$ are in A .
 - Ex. Cut at $\frac{1}{2}$. We must have $\frac{1}{2} \in B$, or else A has a largest element
 - In general, the value you cut at should be in B .
 - Can be expressed as $(-\infty, \frac{1}{2})_{\mathbb{Q}} \sqcup [\frac{1}{2}, \infty)_{\mathbb{Q}}$.
 - $\sqcup$ is the union between 2 disjoint sets
 - What does $A \mid B < C \mid D$ mean?
 - $A \subsetneq C$
 - Pugh chooses not to use $X \subseteq Y$, as most of the time it doesn't matter whether or not $X = Y$. Instead, when $X \neq Y$, he will explicitly say that, using $\subsetneq$.
 - Clarification on collections vs. sets
 - Collection is a set of sets (should be equivalent to family), whereas a set contains elements
 - Can think of $\mathbb{R}$ as the collection of all Dedekind cuts
 - $\mathbb{R}$ is a collection of pairs of subsets.
- Using Dedekind Cuts for the LUB property of $\mathbb{R}$
 - Looking at a subset S of $\mathbb{R}$.
 - S is a nonempty collection of Dedekind cuts.
 - By definition of LUB property, we assume that S is bounded above. $\exists X \mid Y$ s.t. $\forall A \mid B \in S$,
 $A \mid B \leq X \mid Y$
 - Find $C \mid D = \text{LUB of } S \subset R$
 - Suppose $A \mid B \in S$, Let $C = \{a \in A : A \mid B \in S\}$. D is automatically defined as $\mathbb{Q} \setminus C = C^C$
 - C contains all elements of the left handed parts of all cuts
 - $C = \cup A \Rightarrow C \geq A$.
 - C is the LUB because removing any elements means that for some A , $C \not\geq A$
 - $C \mid D$ is a cut built from S
- Proving properties of a field
 - Addition: $A \mid B + C \mid D = E \mid F$
 - $E = \{r \in \mathbb{Q} : \exists a \in A, c \in C (r = a + c)\}$

- $E = A + C$
- Addition is commutative
- Multiplication
 - Try to base on multiplication first
 - $E = \{r \in \mathbb{Q} : \exists a \in A, c \in C(r = a \cdot c)\}$
 - Wrong! Signs mess things up, since two negative values multiply to a positive value.
 - Must do casework instead
 - Start with 2 positive cuts
 - Define F first, and E will be the complement to deal with negative values.
 - $F = \{r \in \mathbb{Q} : \exists b \in B, d \in D(r = b \cdot d)\}$
 - 2 Negative cuts.
 - Can use the corresponding positive cuts.
 - 1 positive 1 negative
 - Flip negative cut, multiply the 2 positive cuts, flip resulting cut back
 - Commutativity and associativity and ugly to prove here, but they work